Machine Learning e Pattern Recognition – Exam Collection

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1 PCA

Motivation. Dimensionality reduction maps data from an n-dimensional space to an m-dimensional one, with $m \ll n$. The practical purposes are four-fold: **to compress information, remove unwanted variability (noise), simplify classification by reducing the number of parameters to estimate, and visualize the data.**

PCA is the standard linear and unsupervised technique for this task. The matrix $P \in \mathbb{R}^{n \times m}$ with orthonormal columns minimizes the average reconstruction error:

$$P^* = \arg\min_P \frac{1}{K} \sum_i \|\mathbf{x}_i - PP^T\mathbf{x}_i\|^2 \quad \text{s.t. } P^T P = I$$

Equivalently, it maximizes the projected variance:

$$\widetilde{L}(P) = \mathrm{Tr}\left(P^T \underbrace{\frac{1}{K} \sum_i \mathbf{x}_i \mathbf{x}_i^T}_{=C} P \right)$$

The solution consists of the m eigenvectors of C with the largest eigenvalues. From the decomposition $C = U\Sigma U^T$:

$$P^* = [\mathbf{u}_1, \ldots, \mathbf{u}_m]$$

The result is a coordinate system where each axis captures the maximum remaining variance, and the variance along each axis corresponds to the associated eigenvalue.

Non-centered data. If the data are not zero-mean, the first eigenvector points towards the dataset mean — not informative. The mean $\bar{\mathbf{x}}$ should always be subtracted, and C becomes the empirical covariance matrix:

$$C = \frac{1}{K} \sum_i (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^T$$

Choice of m. m is a hyperparameter: it does not emerge from the PCA solution itself. Two approaches: cross-validation, or a criterion on explained variance — choose the smallest m such that:

$$\frac{\sum_{i=1}^{m} \sigma_i}{\sum_{i=1}^{n} \sigma_i} \geq t \quad \text{(e.g. } t = 0.95\text{)}$$

The choice of m introduces a trade-off. If m is too large, too many dimensions remain, increasing the number of parameters to estimate in the downstream classifier and the risk of overfitting. If m is too small, discriminant directions may be lost, causing underfitting. Cross-validation allows balancing this trade-off for the specific task.

Limitations. Computing C is expensive in high-dimensional spaces. Most importantly: being unsupervised, PCA does not guarantee to preserve discriminant directions — directions of highest variance do not necessarily coincide with those useful for class separation. **A further practical advantage is that PCA automatically eliminates directions with zero or very low variance, which could cause numerical instability in some classifiers** (e.g., division by zero variance).

2 LDA

Motivation. LDA is a linear **supervised** technique for dimensionality reduction and classification. **Unlike PCA, which maximizes variance without knowledge of class labels, LDA exploits class information to find directions that explicitly separate groups.** The goal is to find the direction $\mathbf{w}$ that maximizes the separation between classes relative to the within-class scatter:

$$\max_{\mathbf{w}} L(\mathbf{w}) = \frac{s_B}{s_W} = \frac{\mathbf{w}^T S_B \mathbf{w}}{\mathbf{w}^T S_W \mathbf{w}}$$

where the scatter matrices are:

$$S_B = \frac{1}{N} \sum_{c=1}^{K} n_c (\boldsymbol{\mu}_c - \boldsymbol{\mu})(\boldsymbol{\mu}_c - \boldsymbol{\mu})^T, \qquad S_W = \frac{1}{N} \sum_{c=1}^{K} \sum_{i=1}^{n_c} (\mathbf{x}_{c,i} - \boldsymbol{\mu}_c)(\mathbf{x}_{c,i} - \boldsymbol{\mu}_c)^T$$

S_B measures how far the class means are from the global mean; S_W measures how spread out the samples are around their class mean. Maximizing the ratio means finding a projection where classes are compact internally and far apart from each other.

Solution derivation. Setting $\nabla_{\mathbf{w}} L = 0$, dividing by the scalar $(\mathbf{w}^T S_W \mathbf{w})$ and multiplying by S_W^{-1}, leads to the eigenvalue problem:

$$S_W^{-1} S_B \mathbf{w} = \lambda(\mathbf{w})\, \mathbf{w}, \qquad \lambda(\mathbf{w}) = L(\mathbf{w})$$

Maximizing L is thus equivalent to maximizing λ: the optimal direction is the eigenvector of $S_W^{-1} S_B$ with the largest eigenvalue.

Binary case. With two classes, $S_B = k(\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1)(\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1)^T$, which has rank 1, and the solution simplifies in closed form:

$$\mathbf{w} \propto S_W^{-1}(\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1)$$

This formula has a direct interpretation: compute the difference between the two class means and "correct" it for within-class scatter via S_W^{-1}. If the classes have the same within-class scatter ($S_W = I$), the optimal direction is simply the one joining the two means.

The classification rule assigns $\mathbf{x}_t$ to the closest class in the projected space, with threshold $t = \frac{m_1 + m_2}{2}$:

$$C(\mathbf{x}_t) = \begin{cases} C_1 & \mathbf{w}^T \mathbf{x}_t < t \\ C_2 & \mathbf{w}^T \mathbf{x}_t \geq t \end{cases}$$

The threshold t is the mean of the projected class means: it is the optimal choice under the assumption of equal within-class variance and equal prior. If the priors differ, the threshold must be shifted accordingly.

Extension to m dimensions. The m most discriminant directions are collected in $W \in \mathbb{R}^{n \times m}$. The criterion becomes:

$$\max_W \ \mathrm{Tr}\left(\tilde{S}_W^{-1} \tilde{S}_B\right), \quad \tilde{S}_B = W^T S_B W,\ \tilde{S}_W = W^T S_W W$$

The solution consists of the m eigenvectors of $S_W^{-1} S_B$ with the largest eigenvalues. **An important structural constraint: the rank of S_B is at most $K-1$, because it is defined by K means constrained to sum to the global mean. As a result, the maximum number of discriminant directions that can be extracted is $K-1$, regardless of the original data dimension. For binary classification, there is always only one direction.**

Limitations and practical considerations. LDA assumes that the within-class distributions are Gaussian with the same covariance matrix — an assumption often violated in practice. S_W can become singular when the number of samples is less than the data dimension, making S_W^{-1} not computable. **A common solution is to apply PCA as a pre-processing step before LDA: PCA reduces dimensionality by eliminating directions with zero or near-zero variance, making S_W invertible and reducing the risk of overfitting. However, care must be taken not to eliminate discriminant directions with PCA before LDA can find them — the number of PCA components should be chosen carefully, typically via cross-validation.**

3 PCA vs LDA

Both **PCA** and **LDA** are linear dimensionality reduction techniques that map data from an n-dimensional feature space to a lower m-dimensional space ($m \ll n$), but they differ fundamentally in supervision, objective, and the nature of the directions they find.

Goals and Formulation

PCA is an *unsupervised* technique. Its goal is to find an orthogonal projection matrix $P \in \mathbb{R}^{n \times m}$ that best preserves the structure of the data, without any knowledge of class labels. Data is first centered by subtracting the mean $\bar{\mathbf{x}}$, and then PCA seeks directions that capture the most variance. The projection of a sample is $\mathbf{y} = P^T \mathbf{x}$, and the reconstruction in the original space is $\hat{\mathbf{x}} = P\mathbf{y}$.

LDA is a *supervised* technique. Its goal is to find a projection matrix $W \in \mathbb{R}^{n \times m}$ such that the projected data has maximum separation between classes

and minimum spread within each class. It explicitly uses class labels to compute class means $\boldsymbol{\mu}_c$ and the global mean $\boldsymbol{\mu}$. The projection of a sample is $\tilde{\mathbf{x}} = W^T\mathbf{x}$.

Training Objective

The PCA objective is to **minimize the average reconstruction error**:

$$P^* = \arg\min_P \frac{1}{K}\sum_{i=1}^{K} \|\mathbf{x}_i - PP^T\mathbf{x}_i\|^2, \quad \text{s.t. } P^TP = I$$

This is equivalent to **maximizing the projected variance**:

$$\widetilde{L}(P) = \mathrm{Tr}\big(P^T C\, P\big)$$

where $C = \frac{1}{K}\sum_i(\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^T$ is the empirical covariance matrix. The constraint $P^TP = I$ ensures orthonormality of the components.

The LDA objective is to **maximize the Fisher ratio** — the ratio of between-class variability to within-class variability. **For the multi-direction case, the criterion is:**

$$L = \mathrm{Tr}\Big(\widetilde{S}_W^{-1}\widetilde{S}_B\Big) = \mathrm{Tr}\big((W^TS_WW)^{-1}(W^TS_BW)\big)$$

where the scatter matrices are:

$$S_B = \frac{1}{N}\sum_{c=1}^{K} n_c(\boldsymbol{\mu}_c - \boldsymbol{\mu})(\boldsymbol{\mu}_c - \boldsymbol{\mu})^T, \quad S_W = \frac{1}{N}\sum_{c=1}^{K}\sum_{i=1}^{n_c}(\mathbf{x}_{c,i} - \boldsymbol{\mu}_c)(\mathbf{x}_{c,i} - \boldsymbol{\mu}_c)^T$$

Unlike PCA, LDA does *not* require W to be orthogonal.

Characteristics of the Directions

The table below summarizes the key properties of the directions found by each method:

Property	PCA Principal Components	LDA Discriminant Directions
Computed from	Eigenvectors of covariance matrix C	Eigenvectors of $S_W^{-1} S_B$
Ordering	By descending eigenvalue (= variance)	By descending eigenvalue (= Fisher ratio)
Orthogonality	Guaranteed ($P^T P = I$)	Not required
Max number of directions	Up to n (feature dimensionality)	At most $K - 1$ (number of classes minus one)
Label-awareness	No — may not be discriminant	Yes — explicitly maximizes class separation
Interpretability	Directions of highest variance	Directions of highest class separability

The $K - 1$ limit on LDA directions follows directly from the rank of S_B: since S_B is a sum of K rank-one matrices centered around the global mean, it has at most $K - 1$ non-zero eigenvalues. In contrast, PCA can extract up to n components, selected by retaining enough to explain a target fraction t of total variance:

$$\min_m \; m \quad \text{s.t.} \quad \frac{\sum_{i=1}^m \sigma_i}{\sum_{i=1}^n \sigma_i} \geq t$$

Use in Classification

PCA for classification is used as a *preprocessing step*. It reduces dimensionality before feeding data into a separate classifier (e.g., logistic regression, k-NN). Because PCA is unsupervised, there is no guarantee the retained components are discriminant — variance-maximizing directions and class-separating directions can differ significantly.

LDA for classification works directly. For a binary problem, the optimal direction is:

$$\mathbf{w} \propto S_W^{-1}(\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1)$$

and the classification rule assigns sample $\mathbf{x}_t$ to class C_1 if $\mathbf{w}^T \mathbf{x}_t < t$, where the threshold is the midpoint of the projected class means:

$$t = \frac{m_1 + m_2}{2}, \quad m_c = \mathbf{w}^T \boldsymbol{\mu}_c$$

This is equivalent to a nearest-centroid classifier in the projected space. LDA implicitly assumes that within-class distributions are Gaussian with equal covariance, so the linear decision boundary is optimal under those assumptions.

A common practical strategy is to **apply PCA first, then LDA**: PCA removes noise and reduces the risk of S_W becoming singular (which happens

when the number of features is large relative to the number of samples), and LDA then finds the most discriminant subspace within the PCA-reduced space.